

E-Commerce Sales Analysis & Business Insights Report

Virtual Data Analytics Internship Project

Submitted by:

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Submitted to:

Primeor Solutions

Tools & Technologies Used:

- Microsoft Excel
- PostgreSQL (pgAdmin)
- Power BI
- SQL
- DAX

Project Modules:

- Data Cleaning & Preparation
- Exploratory Data Analysis
- SQL Business Analysis
- Data Modelling in Power BI
- Interactive Dashboard Development
- Business Insights & Recommendations

Submission Date:

5th June 2026

Internship Duration:

1 Month Virtual Internship [15/05/26] - [15/06/26]

Executive Summary

This project presents a comprehensive analysis of an E-Commerce sales dataset as part of a Virtual Data Analytics Internship at **Primeor Solutions**. The primary objective of the project was to analyze business performance, identify sales and profitability trends, and generate actionable business insights using modern analytics tools and techniques.

The project involved multiple stages including data cleaning and preparation in Microsoft Excel, basic analysis and Pivot tables in Excel, business query analysis using SQL, and interactive dashboard development using Power BI. The dataset had some missing values, duplicate records, and unstandardize data formats which were treated and handled using data cleaning to ensure accurate analysis.

Key performance indicators such as Total Sales, Total Profit, Profit Margin, Average Order Value, Customer Segmentation, and Regional Sales Distribution were analyzed to evaluate overall business performance. Interactive dashboards were developed to visualize sales trends, category-wise profitability, top products and customers, market performance, and the impact of discounting on profitability.

The analysis revealed that certain markets and product categories contributed significantly to overall revenue, while excessive discounting negatively affected profit margins in several sub-categories. Additionally, customer segment analysis highlighted the dominant contribution of consumer-based sales to total business revenue.

The final outcome of this project provides meaningful business insights and recommendations that can support data-driven decision-making, improve profitability, optimize discount strategies, and enhance overall business performance.

Project Objective

The objective of this project was to perform **end-to-end business analysis** on an E-Commerce sales dataset in order to identify key sales trends, profitability patterns, customer behavior, and market performance. The project aimed to transform raw transactional data into meaningful business insights using data analytics techniques and visualization tools.

The analysis focused on evaluating important business metrics such as revenue generation, profit performance, regional sales distribution, customer segmentation, product category performance, and the impact of discounting on profitability.

The project was designed to simulate a real-world business intelligence workflow involving:

- Data cleaning and pre-processing.
- Exploratory Data Analysis (EDA).
- SQL-based Analysis.
- Data Modeling and Dashboard development in Power BI
- Insights generation and Business recommendations.

The final goal of the project was to support data-driven decision-making through analytical reporting and interactive visual dashboards.

Dataset Overview

The dataset used for this project contains E-Commerce transactional sales data in rows and columns including customer orders, product details, shipping information, regional markets, and profitability metrics. The dataset was cleaned, processed, analyzed and visualized to understand overall business performance and generate strategic business insights.

Dataset Summary

Attribute	Details
Total Records	51,325
Total Columns	21
Duplicate Records Removed	54
Data Type	Structured Tabular Dataset
Primary Tools Used	Excel, pgAdmin, Power BI
Analysis Type	Sales & Business Performance Analysis

Key Columns in Dataset

The dataset includes the following important business attributes:

- Order ID
- Order Date
- Ship Date
- Ship mode
- Customer Name
- Segment
- State
- Country
- Market
- Region
- Product id
- Product name
- Product Category

- Sub-Category
- Sales
- Quantity
- Discount
- Profit
- Shipping Cost
- Order Priority

Business Relevance of Dataset

The dataset provides valuable information for analyzing:

- Revenue generation across markets and regions.
- Customer segmentation and behavior analysis.
- Product profitability.
- Impact of discounts on profit margins.
- Shipping performance and operational trends.

This data enables businesses to identify growth opportunities, optimize pricing strategies, and improve overall operational efficiency.

Data Cleaning & Preparation

Before performing analysis and visualization, the dataset was cleaned and prepared to ensure data accuracy, consistency, and reliability. Data preprocessing was carried out primarily using Microsoft Excel and later validated during SQL and Power BI integration.

Cleaning Log :

Steps	Issues Identified	Action Taken	Result
Duplicate Removal	54 duplicate rows found	Removed duplicate rows using Excel Remove Duplicates tool	54 duplicate rows removed
Missing Value Handling	Missing customer names	Replaced blank values with "Unknown Customer"	Missing names handled successfully
Profit Data Correction	Missing profit values	Replaced 14 blank profit values with 0 where rows were otherwise complete	Profit column standardized
Date Format Standardization	Inconsistent date formats	Converted dates into DD-MM-YYYY format using Text to Columns	Dates standardized
Numeric Formatting Standardization	Formatting inconsistencies in numeric columns	Applied numeric formatting with decimal places	Numeric fields standardized
Text Cleaning & Formatting	Extra spaces and text inconsistencies	Cleaned spaces using TRIM / Find & Replace	Text formatting improved

Excel Analysis & Exploratory Data Analysis (EDA)

After cleaning and preparing the dataset, exploratory data analysis was performed using **Microsoft Excel** to identify key business trends, sales patterns, and profitability insights. Pivot tables, KPI calculations, filters, slicers, and charts were used to summarize and analyze the data effectively.

Key Performance Indicators (KPIs)

- Total Sales
- Total Profit
- Average Discount
- Total Orders
- Total Customers

Pivot table Analyze

- Region wise sales performance
- Category wise profit analysis
- Segment wise sales contribution
- Monthly sales trend
- Top performing products

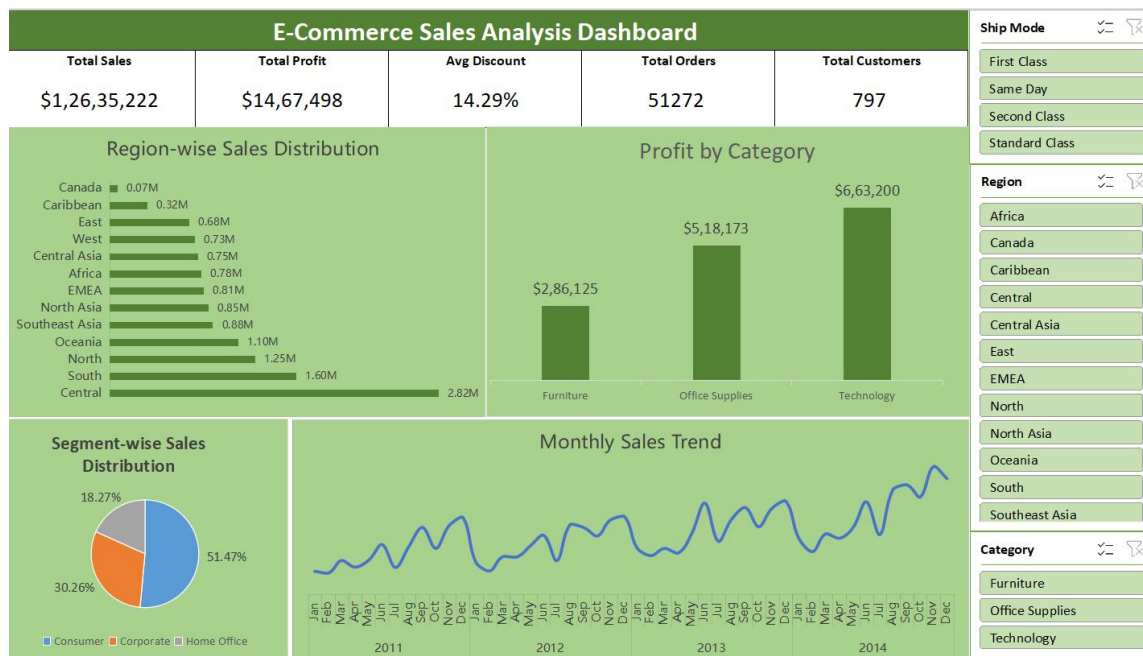


Image : Excel Sales Analysis Dashboard

SQL Analysis & Business Query Insights

After completing data cleaning and Excel analysis, SQL analysis was performed using **PostgreSQL (pgAdmin)** to extract meaningful business insights from the E-Commerce dataset. SQL queries were used to analyze sales performance, profitability trends, customer behavior, product performance, and operational efficiency. The cleaned dataset was imported into PostgreSQL, where multiple analytical queries were executed to answer important business questions.

Objectives of SQL Analysis

- Identifying top products and customers.
- Analyzing Region and Category wise sales
- Analyzing customer and product performance
- Measuring sales contribution across segments
- Customer Segmentation
- Category and Sub-category analysis

Below are some of the Queries performed

```
3  --Top 10 profitable products
4  select product_name, round(sum(profit),2) as total_profit
5  from ecommerce_data
6  group by product_name
7  order by total_profit desc
8  limit 10;
```

Output

	product_name text	total_profit numeric
1	Canon imageCLASS 2200 Advanced Copier	25200.00
2	Cisco Smart Phone, Full Size	17241.00
3	Motorola Smart Phone, Full Size	17028.00
4	Hoover Stove, Red	11806.00
5	Sauder Classic Bookcase, Traditional	10671.00
6	Harbour Creations Executive Leather Armchair, Adjusta...	10429.00
7	Nokia Smart Phone, Full Size	9934.00
8	Cisco Smart Phone, with Caller ID	9783.00
9	Nokia Smart Phone, with Caller ID	9465.00
10	Belkin Router, USB	8951.00

Query for category wise average profit

```
29  --Category-wise average profit
30  select category, round(avg(profit),2) as avg_profit
31  from ecommerce_data
32  group by category
33  order by avg_profit desc;
```

Output

	category text	avg_profit numeric
1	Technology	65.42
2	Furniture	29.01
3	Office Supplies	16.58

Query for average discount per category

```
38  select
39      category,
40      concat(round(avg(discount) * 100, 2), '%') as avg_discount_per
41  from ecommerce_data
42  group by category
43  order by avg_discount_per desc;
```

Output

	category text	avg_discount_per text
1	Furniture	16.80%
2	Office Supplies	13.74%
3	Technology	13.53%

Major Business Insights from SQL Analysis :

- **APAC** generated the highest revenue among all markets, making it the company's strongest performing region.
- The **Technology** category generated the highest overall profit, indicating strong profitability and higher-value product sales.
- **Machines, Phones**, and certain Furniture products contributed significantly to negative profits.
- **Standard Class** was the most frequently used shipping mode by Customers.
- **South, North**, and **Central** regions contributed highly to total sales, while **Caribbean** and **Canada** showed comparatively weaker sales performance.

Power BI Dashboard Analysis

After completing data cleaning and preparation with Excel and SQL analysis, interactive dashboards were developed using **Microsoft Power BI** to visualize business performance and generate actionable insights. A **star schema data model** was created to improve analytical efficiency and support dynamic filtering across reports.

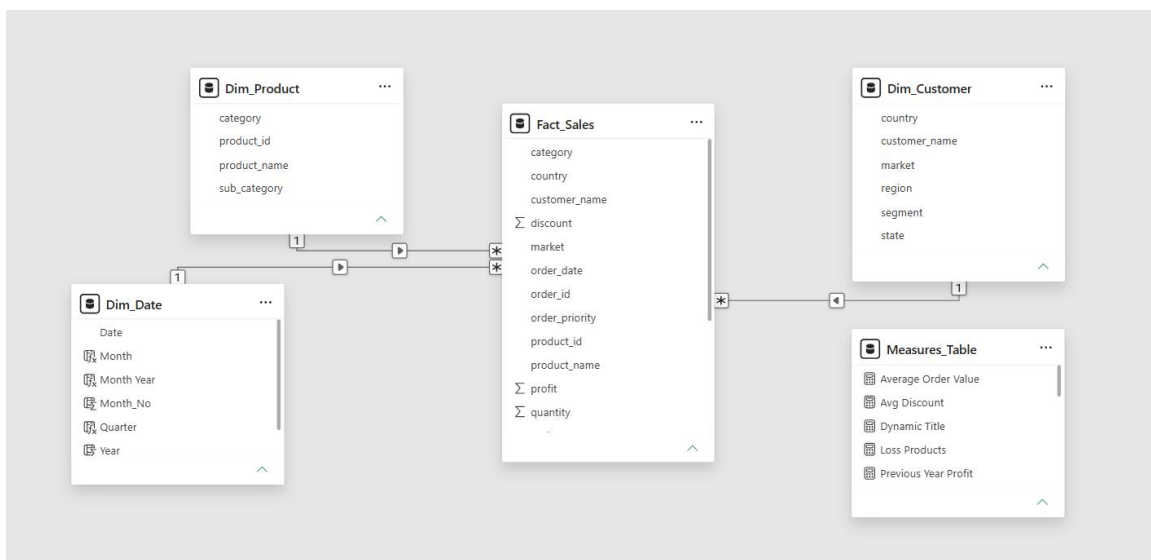
Power BI was used to:

- Extract Transform and Load (ETL) using Power Query.
- Data Modelling - Star Schema was built for proper filtering.
- Creating DAX measures and KPIs.
- Visualize sales, product and customer data using charts.
- Generate decision driven business insights.

Data Modelling (Star Schema) :

The whole E commerce dataset after loading from power query was modeled in Power BI. Star Schema architecture was used to modeled data in fact and dimension tables. Also appropriate relationships were built between primary and foreign keys. Following tables were built using star schema,

- Fact_Sales
- Dim_Product
- Dim_Customer
- Dim_Date



Data Modelling usin Star Schema

A. Executive Overview Dashboard

The Executive Overview Dashboard was designed to provide a high-level summary of overall business performance. This dashboard focused on key financial and operational metrics to help stakeholders quickly understand revenue trends, profitability, customer activity, and regional performance.

Key KPIs Included

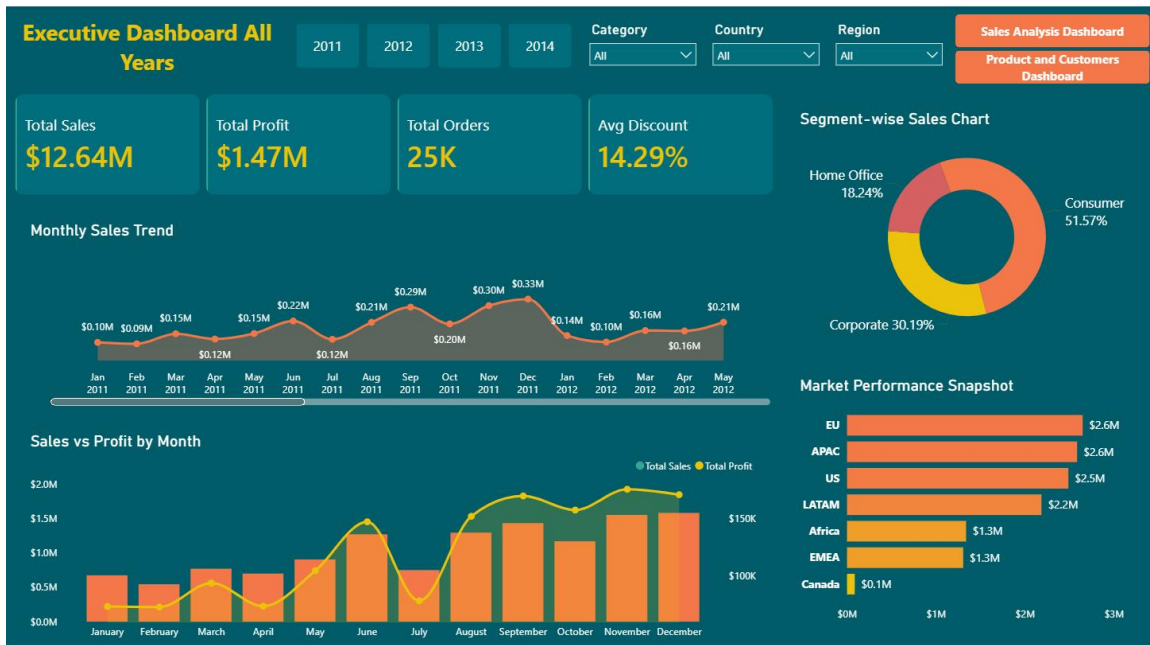
- Total Sales
- Total Profit
- Total Orders
- Average Discount

Major Visualizations

- Monthly Sales Trend (Line Chart)
- Market Sales Distribution (Bar Chart)
- Sales vs Profit Comparison (Line and Column Chart)
- Segment-wise Sales Contribution (Donut Chart)
- Date Buttons, Category, Country and Market Slicers

Key Business Insights

- Total sales reached **\$12.64M** with an overall profit of **\$1.47M** generated from **25K** orders.
- The average discount across all orders is **14.29%**, indicating frequent promotional pricing strategies.
- Consumer segment contributes the highest share (**51.57%**), followed by Corporate and Home Office customers.
- **EU** and **APAC** regions are the top-performing markets with sales **\$2.6M**, while Canada shows the lowest contribution.
- Monthly sales show a generally upward trend, with strong growth observed in **September** and year-end months.
- Profit performance closely follows sales growth, indicating healthy revenue generation during high-sales periods.



Dashboard 1 : Executive Overview Dashboard in Power BI

Key Features of Dashboard

- Consistent Colour Theme used.
- Modern and Minimal Design.
- Dynamic Title.
- Slicers for dynamic filtering.
- Insights text box.

B. Sales Analysis Dashboard

The Sales Analysis Dashboard focused on detailed sales and profitability analysis across different regions, markets, categories, and customer segments. The objective of this dashboard was to identify revenue drivers, profitability trends, and the impact of discounting on business performance.

Key KPIs Included

- Total Sales
- Total Profit
- Profit Margin %
- Average Discount
- Average Order Value
- PY Sales and PY Profit
- YOY Growth %

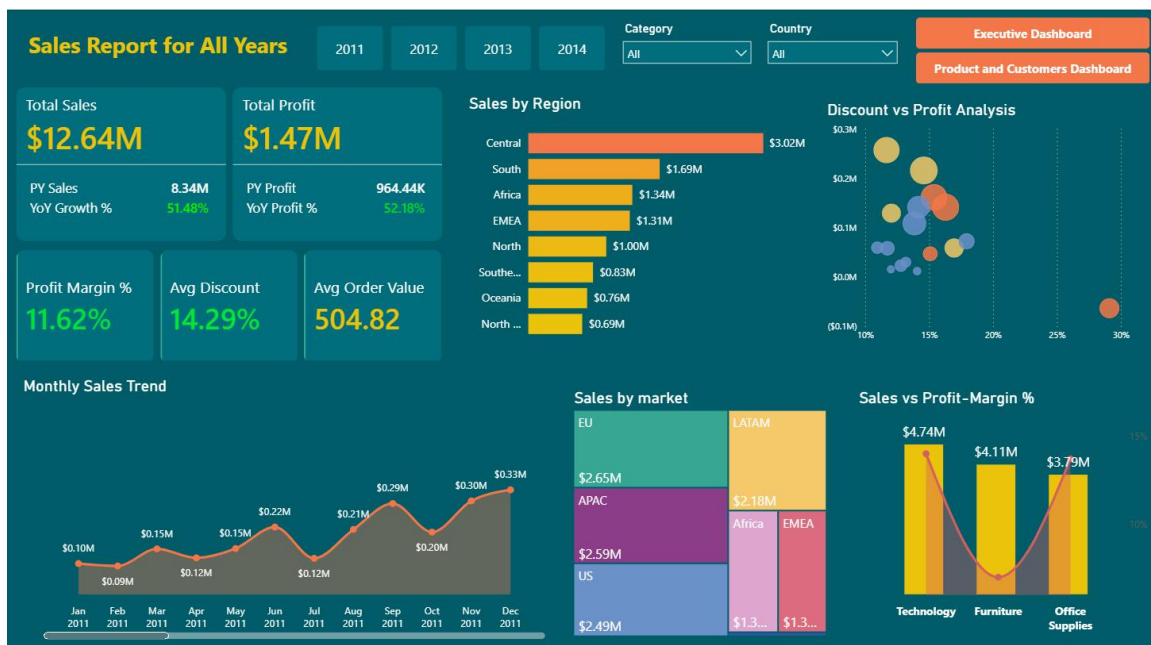
Major Visualizations

- Monthly Sales Trend (Line Chart)
- Regional Sales Distribution (Bar Chart)
- Market Sales Distribution (Treemap)
- Sales vs Profit Comparison by Category (Clustered Column Chart)
- Discount vs Profit Analysis (Scatter Plot)
- Date Buttons, Category, Country and Market Slicers

Key Business Insights

- The business maintains a healthy profitability between **11.00% - 12.00%** profit margin through all the years.
- Average Order Value is **504.82**, suggesting customers place relatively high-value orders.
- **Central** Region generates the highest sales **\$3.02M**, making it highest performing region followed by **South** region **\$1.69M**. Whereas **North** region contributes less **\$0.69M** sales which may need proper expansion strategies.
- **EU** market contributes higher revenue **\$2.65M** followed by **APAC** and **US**, whereas **Africa**, **EMEA** and **Canada** are least contributing markets.
- Monthly sales trend shows upward trend in month-ends specifically from September to December.

- **Technology** and **Office Supplies** maintains balanced sales and profitability, whereas **Furniture** has strong sales but significantly lower profit, indicating lower margins or excessive discounts.
- Some products like **Phones** and **Copiers** remain profitable even with moderate discounts around **11.00%** . While products like **Tables** with higher discounts around **30,00%** reduce profitability and some times leads to negative profit.



Dashboard 2 : Sales Analysis Dashboard in Power BI

Key Features of Dashboard

- Consistent Colour Theme used.
- Modern and Minimal Design.
- Dynamic Title.
- Slicers for dynamic filtering.
- Dynamic KPIs with reference labels.
- Values with Conditional Formatting.

C. Product & Customer Insights Dashboard

The Product & Customer Insights Dashboard was developed to analyze top products and customers, customer behavior, product performance, and profitability at a detailed level. This dashboard focused on identifying top-performing products, major customers, and loss-making products and sub-categories in business areas. Also it provides sales and profit comparison of sub-categories.

Key KPIs Included

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Profit Margin %
- Average Discount
- Average Order Value
- PY Sales and PY Profit Margin %

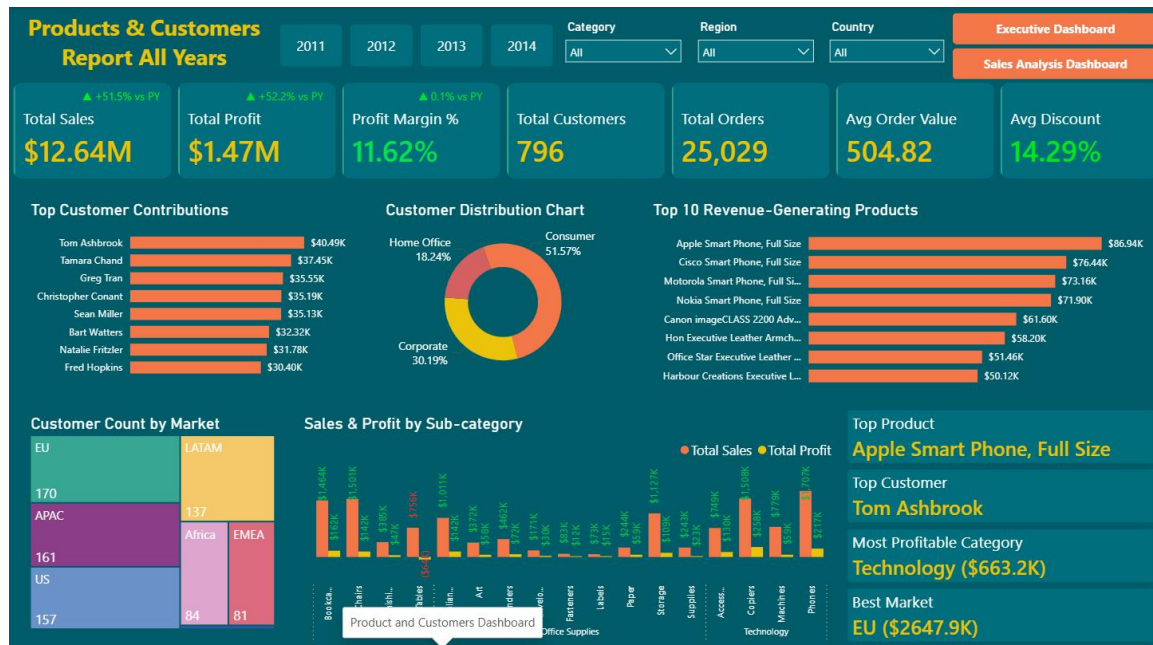
Major Visualizations

- Top Customers Contribution (Bar Chart)
- Top 10 Revenue - generating products (Bar Chart)
- Customer Segment Distribution (Donut Chart)
- Sales vs Profit Comparison by Sub-Category (Clustered Column Chart)
- Date Buttons, Category, Country and Market Slicers
- Insights Box for quick business insights.

Key Business Insights

- **Phones, Copiers, and Chairs** are among the most profitable sub-categories, while **Tables** generated negative profit despite strong sales mainly due to excessive discounts.
- High performing products are dominated by **Phones** and **Copiers** each generating **\$1.71M** and **\$1.51M** sales, showing strong market demand in **Technology** Category.
- The average discount stands at **14.29%**, suggesting discounting plays a major role in driving sales volume.
- The **Consumer** segment contributes the highest share of customers (**51.57 %**), making it the primary revenue-driving segment.

- The top performing customers contributes nearly from **\$37K to \$40K** sales making them top performers.
- Total Customers and Orders grow gradually during the years maintaining profit margin % between **11.00% to 12.00%**.



Dashboard 3 : Products & Customers Analysis Dashboard in Power BI

Key Features of Dashboard

- Consistent Colour Theme used.
- Modern and Minimal Design.
- Dynamic Title.
- Slicers for dynamic filtering.
- Insights text box.
- Dynamic KPIs with reference labels.
- Values with Conditional Formatting.

Key Business Recommendations

Based on the overall analysis conducted using Excel, SQL, and Power BI dashboards, the following business recommendations are suggested to improve sales performance, profitability, operational efficiency, and long-term business growth.

1. Optimize Discount Strategies

The analysis revealed that excessive discounting negatively impacts profitability, particularly within low-margin sub-categories such as **Tables**. While moderate discounts support sales growth, higher discount percentages often lead to reduced or negative profit margins.

Recommended Actions:

- Implement controlled discount policies for low-profit products.
- Apply data-driven pricing strategies based on product profitability.
- Monitor high-discount products regularly to avoid revenue leakage.

2. Improve Profitability in Furniture Category

Although the **Furniture** category generates strong sales revenue, its profitability remains significantly lower compared to **Technology** and **Office Supplies**.

Recommended Actions:

- Re-evaluate pricing structure for Furniture products.
- Reduce excessive discounts within Furniture sub-categories.
- Focus on promoting high-margin furniture items.

3. Expand Low-Performing Regions & Markets

The analysis identified that regions such as **North** and markets including **Africa**, **EMEA**, and **Canada** contribute comparatively lower sales revenue.

Recommended Actions:

- Develop region-specific marketing and expansion strategies.
- Improve customer acquisition campaigns in under-performing markets.
- Increase promotional activities and strategic partnerships in low-performing regions.

4. Strengthen High-Performing Markets

EU, APAC, and Central regions consistently generate the highest sales and revenue contribution, making them critical growth drivers for the business.

Recommended Actions:

- Continue investment in high-performing markets.
- Introduce customer loyalty programs and retention strategies.
- Expand product availability and premium offerings in strong-performing regions.
- Use successful strategies from top markets as benchmarks for other regions.

5. Focus on High-Profit Product Categories

Technology products, particularly **Phones** and **Copiers**, contribute significantly to both revenue and profitability, indicating strong customer demand and healthy margins.

Recommended Actions:

- Prioritize inventory and marketing for high-performing technology products.
- Increase cross-selling and up-selling opportunities within profitable categories.
- Launch targeted campaigns for top-selling products.
- Monitor demand trends to maintain competitive market positioning.
- Keep moderate discounts across Technology category products.

6. Enhance Customer Retention Strategies

The **Consumer segment** represents the largest share of total customers and sales contribution, making it the most valuable customer group.

Recommended Actions:

- Introduce personalized promotions and loyalty programs for consumer customers.
- Analyze purchasing patterns to improve customer engagement.
- Develop retention strategies for high-value customers.
- Use customer segmentation for targeted marketing campaigns.

7. Monitor Seasonal Sales Trends

Monthly sales trends indicate stronger business performance during **September** to **December**, suggesting seasonal demand fluctuations.

Recommended Actions:

- Increase inventory planning before peak sales periods.
- Launch seasonal promotional campaigns during high-demand months.
- Improve operational readiness for year-end sales spikes.

8. Improve Loss-Making Product Performance

Certain products continue to generate negative profit despite maintaining strong sales volume like **Tables**.

Recommended Actions:

- Review pricing and operational costs for loss-making products.
- Reduce discounts on consistently unprofitable items.

Overall Business Conclusion

The analysis indicates that the business maintains strong revenue generation of **\$12.64M** and stable profitability margin of **11.50%** across major markets. However, opportunities exist to further improve profit margins of products in **Furniture** category through optimized discount strategies, improved regional expansion, and better management of low-performing products and categories.

Technology emerged as highest performing category as well as markets and regions like **EU, APAC** and **Central** respectively contributes largely in sales and profitability.

Customers from **Consumer** segment contributes more revenue as compared to others making it most valuable customer group.

Also most of the customers prefer **Standard shipping mode** over premium and high priced shipping modes.

By focusing on profitable markets, strengthening customer retention, and implementing data-driven pricing and operational strategies, the business can achieve sustainable growth and improved long-term profitability.

Conclusion

This project successfully demonstrated an **End-to-End Business Analytics** workflow using Excel, SQL, and Power BI to analyze **E-Commerce sales data** and generate actionable business insights.

The analysis process included data cleaning, preprocessing, exploratory analysis, SQL-based business querying, dashboard development, and insights generation. Interactive dashboards and analytical visualizations helped identify important trends related to sales performance, profitability, customer behavior, regional contribution, and product performance.

The project also highlighted the importance of **data-driven decision-making** in improving pricing strategies, optimizing profitability, and identifying growth opportunities across regions and product categories.

Overall, this project provided valuable practical experience in:

- Data cleaning and preparation
- SQL-based analysis
- Data modeling and DAX calculations
- Business intelligence dashboard development
- Business insight generation and reporting

The final dashboards and business insights report can support management in making informed strategic decisions and improving overall business performance.